

```
# Add interaction between predictors and Mlax_Position
> survey_model_interaction <- svyglm(`Wellbeing Score` ~ Overall_Stress_Num *
Mlax_Position +
+           Physical_Fatigue_Num * Mlax_Position +
+           Soreness_Num * Mlax_Position +
+           Sleep_Quality_Num * Mlax_Position +
+           Motivation_Num * Mlax_Position,
+           design = boot_survey_design, family =
gaussian())
> # Display summary of interaction model
> summary(survey_model_interaction)
```

Call:

```
svyglm(formula = `Wellbeing Score` ~ Overall_Stress_Num * Mlax_Position +
Physical_Fatigue_Num * Mlax_Position + Soreness_Num * Mlax_Position +
Sleep_Quality_Num * Mlax_Position + Motivation_Num * Mlax_Position,
design = boot_survey_design, family = gaussian())
```

Survey design:

```
as.svrepdesign.default(final_survey_design, type = "bootstrap",
replicates = 1000)
```

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.824e-13	4.086e-14	-4.465e+00	8.97e-06
Overall_Stress_Num	1.000e+00	3.575e-15	2.797e+14	< 2e-16
Mlax_PositionOM	2.606e-15	8.028e-15	3.250e-01	0.745561
Mlax_PositionDM	-8.700e-15	7.985e-15	-1.090e+00	0.276193
Mlax_PositionD	-1.092e-14	1.827e-14	-5.970e-01	0.550323
Mlax_PositionGK	-9.980e-15	1.112e-14	-8.980e-01	0.369647
Mlax_PositionFO	-2.463e-14	2.155e-14	-1.143e+00	0.253303
Physical_Fatigue_Num	1.000e+00	2.878e-15	3.474e+14	< 2e-16
Soreness_Num	1.000e+00	1.835e-15	5.451e+14	< 2e-16
Sleep_Quality_Num	1.000e+00	1.676e-15	5.967e+14	< 2e-16
Motivation_Num	1.000e+00	1.381e-15	7.243e+14	< 2e-16
Overall_Stress_Num:Mlax_PositionOM	-9.098e-16	4.045e-16	-2.249e+00	0.024737
Overall_Stress_Num:Mlax_PositionDM	7.563e-15	2.211e-15	3.420e+00	0.000652
Overall_Stress_Num:Mlax_PositionD	-6.656e-16	8.160e-16	-8.160e-01	0.414892
Overall_Stress_Num:Mlax_PositionGK	-1.098e-15	6.469e-16	-1.697e+00	0.090032
Overall_Stress_Num:Mlax_PositionFO	-5.988e-16	9.269e-16	-6.460e-01	0.518380
Mlax_PositionOM:Physical_Fatigue_Num	-8.501e-17	3.148e-16	-2.700e-01	0.787195
Mlax_PositionDM:Physical_Fatigue_Num	-1.523e-14	2.061e-15	-7.390e+00	3.17e-13
Mlax_PositionD:Physical_Fatigue_Num	-2.315e-16	6.020e-16	-3.840e-01	0.700712
Mlax_PositionGK:Physical_Fatigue_Num	-3.450e-16	5.141e-16	-6.710e-01	0.502316
Mlax_PositionFO:Physical_Fatigue_Num	-1.328e-15	1.019e-15	-1.303e+00	0.192962
Mlax_PositionOM:Soreness_Num	1.413e-16	3.267e-16	4.330e-01	0.665362
Mlax_PositionDM:Soreness_Num	2.145e-14	3.402e-15	6.303e+00	4.43e-10
Mlax_PositionD:Soreness_Num	4.200e-16	2.487e-15	1.690e-01	0.865891

Mlax_PositionGK:Soreness_Num	2.575e-16	4.875e-16	5.280e-01	0.597517
Mlax_PositionFO:Soreness_Num	3.658e-15	3.696e-15	9.900e-01	0.322633
Mlax_PositionOM:Sleep_Quality_Num	2.802e-16	3.226e-16	8.690e-01	0.385302
Mlax_PositionDM:Sleep_Quality_Num	-1.800e-15	6.320e-16	-2.848e+00	0.004493
Mlax_PositionD:Sleep_Quality_Num	5.973e-16	4.646e-16	1.285e+00	0.198950
Mlax_PositionGK:Sleep_Quality_Num	3.511e-16	4.696e-16	7.480e-01	0.454826
Mlax_PositionFO:Sleep_Quality_Num	-2.087e-15	2.761e-15	-7.560e-01	0.449868
Mlax_PositionOM:Motivation_Num	2.332e-17	3.287e-16	7.100e-02	0.943458
Mlax_PositionDM:Motivation_Num	-1.278e-14	1.994e-15	-6.410e+00	2.27e-10
Mlax_PositionD:Motivation_Num	1.010e-16	5.555e-16	1.820e-01	0.855839
Mlax_PositionGK:Motivation_Num	2.705e-16	5.102e-16	5.300e-01	0.596051
Mlax_PositionFO:Motivation_Num	3.801e-15	3.123e-15	1.217e+00	0.223888

(Intercept)	***
Overall_Stress_Num	***
Mlax_PositionOM	
Mlax_PositionDM	
Mlax_PositionD	
Mlax_PositionGK	
Mlax_PositionFO	
Physical_Fatigue_Num	***
Soreness_Num	***
Sleep_Quality_Num	***
Motivation_Num	***
Overall_Stress_Num:Mlax_PositionOM	*
Overall_Stress_Num:Mlax_PositionDM	***
Overall_Stress_Num:Mlax_PositionD	
Overall_Stress_Num:Mlax_PositionGK	.
Overall_Stress_Num:Mlax_PositionFO	
Mlax_PositionOM:Physical_Fatigue_Num	
Mlax_PositionDM:Physical_Fatigue_Num	***
Mlax_PositionD:Physical_Fatigue_Num	
Mlax_PositionGK:Physical_Fatigue_Num	
Mlax_PositionFO:Physical_Fatigue_Num	
Mlax_PositionOM:Soreness_Num	
Mlax_PositionDM:Soreness_Num	***
Mlax_PositionD:Soreness_Num	
Mlax_PositionGK:Soreness_Num	
Mlax_PositionFO:Soreness_Num	
Mlax_PositionOM:Sleep_Quality_Num	
Mlax_PositionDM:Sleep_Quality_Num	**
Mlax_PositionD:Sleep_Quality_Num	
Mlax_PositionGK:Sleep_Quality_Num	
Mlax_PositionFO:Sleep_Quality_Num	
Mlax_PositionOM:Motivation_Num	
Mlax_PositionDM:Motivation_Num	***
Mlax_PositionD:Motivation_Num	
Mlax_PositionGK:Motivation_Num	

Mlax_PositionFO:Motivation_Num

Signif. codes: 0 '***' 0.001 '**' 0.01 '.' 0.1 ' ' 1

(Dispersion parameter for gaussian family taken to be 7.3323e-26)

Number of Fisher Scoring iterations: 1

```
> model_coefficients_interaction <- coef(summary(survey_model_interaction))
```

```
> conf_intervals_interaction <- confint(survey_model_interaction)
```

```
> print("Survey-Weighted Interaction Model Coefficients:")
```

```
[1] "Survey-Weighted Interaction Model Coefficients:"
```

```
> print(model_coefficients_interaction)
```

	Estimate	Std. Error	t value
(Intercept)	-1.824341e-13	4.086262e-14	-4.464573e+00
Overall_Stress_Num	1.000000e+00	3.575322e-15	2.796951e+14
Mlax_PositionOM	2.605920e-15	8.028309e-15	3.245914e-01
Mlax_PositionDM	-8.700061e-15	7.985164e-15	-1.089528e+00
Mlax_PositionD	-1.091619e-14	1.827021e-14	-5.974860e-01
Mlax_PositionGK	-9.980181e-15	1.111931e-14	-8.975544e-01
Mlax_PositionFO	-2.463173e-14	2.154919e-14	-1.143047e+00
Physical_Fatigue_Num	1.000000e+00	2.878463e-15	3.474076e+14
Soreness_Num	1.000000e+00	1.834518e-15	5.451024e+14
Sleep_Quality_Num	1.000000e+00	1.675790e-15	5.967334e+14
Motivation_Num	1.000000e+00	1.380608e-15	7.243184e+14
Overall_Stress_Num:Mlax_PositionOM	-9.098034e-16	4.045349e-16	-2.249011e+00
Overall_Stress_Num:Mlax_PositionDM	7.562731e-15	2.211091e-15	3.420362e+00
Overall_Stress_Num:Mlax_PositionD	-6.655898e-16	8.160074e-16	-8.156663e-01
Overall_Stress_Num:Mlax_PositionGK	-1.097783e-15	6.469223e-16	-1.696931e+00
Overall_Stress_Num:Mlax_PositionFO	-5.988452e-16	9.268880e-16	-6.460816e-01
Mlax_PositionOM:Physical_Fatigue_Num	-8.501200e-17	3.148246e-16	-2.700297e-01
Mlax_PositionDM:Physical_Fatigue_Num	-1.522941e-14	2.060706e-15	-7.390383e+00
Mlax_PositionD:Physical_Fatigue_Num	-2.314620e-16	6.020236e-16	-3.844733e-01
Mlax_PositionGK:Physical_Fatigue_Num	-3.449873e-16	5.140609e-16	-6.711019e-01
Mlax_PositionFO:Physical_Fatigue_Num	-1.327693e-15	1.019124e-15	-1.302778e+00
Mlax_PositionOM:Soreness_Num	1.413533e-16	3.267110e-16	4.326554e-01
Mlax_PositionDM:Soreness_Num	2.144676e-14	3.402496e-15	6.303243e+00
Mlax_PositionD:Soreness_Num	4.200314e-16	2.486503e-15	1.689245e-01
Mlax_PositionGK:Soreness_Num	2.574502e-16	4.874570e-16	5.281496e-01
Mlax_PositionFO:Soreness_Num	3.657640e-15	3.696194e-15	9.895692e-01
Mlax_PositionOM:Sleep_Quality_Num	2.801957e-16	3.225964e-16	8.685644e-01
Mlax_PositionDM:Sleep_Quality_Num	-1.799997e-15	6.320249e-16	-2.847984e+00
Mlax_PositionD:Sleep_Quality_Num	5.972760e-16	4.646498e-16	1.285433e+00
Mlax_PositionGK:Sleep_Quality_Num	3.511463e-16	4.696375e-16	7.476965e-01
Mlax_PositionFO:Sleep_Quality_Num	-2.087382e-15	2.761298e-15	-7.559423e-01
Mlax_PositionOM:Motivation_Num	2.331963e-17	3.287108e-16	7.094268e-02
Mlax_PositionDM:Motivation_Num	-1.277993e-14	1.993763e-15	-6.409953e+00
Mlax_PositionD:Motivation_Num	1.009544e-16	5.555409e-16	1.817227e-01

Mlax_PositionGK:Motivation_Num	2.705402e-16	5.101984e-16	5.302647e-01
Mlax_PositionFO:Motivation_Num	3.801024e-15	3.123181e-15	1.217036e+00

Pr(>|t|)

(Intercept)	8.970854e-06
Overall_Stress_Num	0.000000e+00
Mlax_PositionOM	7.455608e-01
Mlax_PositionDM	2.761933e-01
Mlax_PositionD	5.503233e-01
Mlax_PositionGK	3.696473e-01
Mlax_PositionFO	2.533029e-01
Physical_Fatigue_Num	0.000000e+00
Soreness_Num	0.000000e+00
Sleep_Quality_Num	0.000000e+00
Motivation_Num	0.000000e+00
Overall_Stress_Num:Mlax_PositionOM	2.473688e-02
Overall_Stress_Num:Mlax_PositionDM	6.516102e-04
Overall_Stress_Num:Mlax_PositionD	4.148925e-01
Overall_Stress_Num:Mlax_PositionGK	9.003248e-02
Overall_Stress_Num:Mlax_PositionFO	5.183803e-01
Mlax_PositionOM:Physical_Fatigue_Num	7.871952e-01
Mlax_PositionDM:Physical_Fatigue_Num	3.171291e-13
Mlax_PositionD:Physical_Fatigue_Num	7.007125e-01
Mlax_PositionGK:Physical_Fatigue_Num	5.023164e-01
Mlax_PositionFO:Physical_Fatigue_Num	1.929615e-01
Mlax_PositionOM:Soreness_Num	6.653619e-01
Mlax_PositionDM:Soreness_Num	4.429323e-10
Mlax_PositionD:Soreness_Num	8.658914e-01
Mlax_PositionGK:Soreness_Num	5.975171e-01
Mlax_PositionFO:Soreness_Num	3.226330e-01
Mlax_PositionOM:Sleep_Quality_Num	3.853016e-01
Mlax_PositionDM:Sleep_Quality_Num	4.493232e-03
Mlax_PositionD:Sleep_Quality_Num	1.989499e-01
Mlax_PositionGK:Sleep_Quality_Num	4.548256e-01
Mlax_PositionFO:Sleep_Quality_Num	4.498684e-01
Mlax_PositionOM:Motivation_Num	9.434581e-01
Mlax_PositionDM:Motivation_Num	2.274715e-10
Mlax_PositionD:Motivation_Num	8.558385e-01
Mlax_PositionGK:Motivation_Num	5.960505e-01
Mlax_PositionFO:Motivation_Num	2.238883e-01

> print("Bootstrapped 95% Confidence Intervals for Interaction Model:")

[1] "Bootstrapped 95% Confidence Intervals for Interaction Model:"

> print(conf_intervals_interaction)

	2.5 %	97.5 %
(Intercept)	-2.626241e-13	-1.022442e-13
Overall_Stress_Num	1.000000e+00	1.000000e+00
Mlax_PositionOM	-1.314906e-14	1.836090e-14
Mlax_PositionDM	-2.437037e-14	6.970246e-15
Mlax_PositionD	-4.677016e-14	2.493777e-14

Mlax_PositionGK	-3.180102e-14	1.184065e-14
Mlax_PositionFO	-6.692046e-14	1.765700e-14
Physical_Fatigue_Num	1.000000e+00	1.000000e+00
Soreness_Num	1.000000e+00	1.000000e+00
Sleep_Quality_Num	1.000000e+00	1.000000e+00
Motivation_Num	1.000000e+00	1.000000e+00
Overall_Stress_Num:Mlax_PositionOM	-1.703674e-15	-1.159328e-16
Overall_Stress_Num:Mlax_PositionDM	3.223625e-15	1.190184e-14
Overall_Stress_Num:Mlax_PositionD	-2.266946e-15	9.357659e-16
Overall_Stress_Num:Mlax_PositionGK	-2.367321e-15	1.717558e-16
Overall_Stress_Num:Mlax_PositionFO	-2.417796e-15	1.220106e-15
Mlax_PositionOM:Physical_Fatigue_Num	-7.028326e-16	5.328086e-16
Mlax_PositionDM:Physical_Fatigue_Num	-1.927339e-14	-1.118542e-14
Mlax_PositionD:Physical_Fatigue_Num	-1.412890e-15	9.499659e-16
Mlax_PositionGK:Physical_Fatigue_Num	-1.353795e-15	6.638203e-16
Mlax_PositionFO:Physical_Fatigue_Num	-3.327650e-15	6.722647e-16
Mlax_PositionOM:Soreness_Num	-4.997935e-16	7.825001e-16
Mlax_PositionDM:Soreness_Num	1.476961e-14	2.812391e-14
Mlax_PositionD:Soreness_Num	-4.459552e-15	5.299615e-15
Mlax_PositionGK:Soreness_Num	-6.991490e-16	1.214050e-15
Mlax_PositionFO:Soreness_Num	-3.595874e-15	1.091115e-14
Mlax_PositionOM:Sleep_Quality_Num	-3.528764e-16	9.132679e-16
Mlax_PositionDM:Sleep_Quality_Num	-3.040300e-15	-5.596933e-16
Mlax_PositionD:Sleep_Quality_Num	-3.145658e-16	1.509118e-15
Mlax_PositionGK:Sleep_Quality_Num	-5.704835e-16	1.272776e-15
Mlax_PositionFO:Sleep_Quality_Num	-7.506231e-15	3.331466e-15
Mlax_PositionOM:Motivation_Num	-6.217517e-16	6.683909e-16
Mlax_PositionDM:Motivation_Num	-1.669255e-14	-8.867312e-15
Mlax_PositionD:Motivation_Num	-9.892545e-16	1.191163e-15
Mlax_PositionGK:Motivation_Num	-7.306874e-16	1.271768e-15
Mlax_PositionFO:Motivation_Num	-2.327993e-15	9.930041e-15

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